Research Statement

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The global Internet depends on a massive, decentralized routing system in which thousands of independently operated networks make continuous control-plane decisions that direct how traffic moves across the world. Ensuring the security of routing is therefore fundamental to preserving the stability, integrity, and resilience of today's Internet.

With today's Internet growing increasingly complex in both topology and business models, the challenges of securing routing remain persistent: **(1) Hard to secure protocols and systems:** existing routing protocols remain vulnerable, and while Resource Public Key Infrastructure (RPKI) was introduced to prevent hijacks, it also introduces new operational risks and attack surfaces. The lack of authoritative global routing information—combined with the complexity of Internet topology and opaque business arrangements—makes it increasingly difficult to distinguish malicious activities from normal configuration changes. **(2) Hard to detect and identify threats:** while adversaries can explore insecure routing systems, detecting such threats is challenging because ground truth about routing intent is invisible. Identifying the responsible party is even harder, as public information is inconsistent and sometimes out-dated, and abnormal threats can hide behind operational configurations and human errors.

My research focuses on the intersection of routing security, data-driven security and Internet measurement. My work build new protocols and systems to address the following problems: **(1)** how to retrieve routing information through active measurement systems and AI-driven data collection pipelines, **(2)** how to apply data-driven insight to improve threat detection, identification, and reasoning, and **(3)** how to design routing systems and protocols that ensure strong security with minimal operational overhead and even in the post-quantum era. Across my projects, I combine active experimentation, multi-plane data collection, and data-driven analysis to reveal hidden routing dependencies, threats, and misconfigurations, enabling more reliable and secure interdomain communication. **My goal is to develop data-driven and AI-assisted approaches that make the global routing ecosystem more observable, more predictable, and fundamentally more secure.**

Past and Current Research

Understanding the deployment and ecosystem of routing security

A fundamental challenge in Internet security is **understanding how the global routing ecosystem actually operates and what its real security posture is**. The core of today's routing security relies on the deployment of RPKI and Route Origin Validation (ROV), which enables networks to identify unauthorized announcements and prevent origin hijacks. Knowing how the Internet is protected and how ROV is deployed across different networks are vital for understanding the real effectiveness of the routing ecosystem, assessing the Internet's resilience to attacks, and guiding operators and policymakers as they adopt and refine routing security practices.

To answer how today's Internet is protected, I built **RoVista** [1], using side-channels to remotely measure the ROV protection for more than 30,000 networks. Instead of just binarily measuring the deployment, RoVista presents a scoring system leveraging large amounts of in-the-wild prefixes to metric the protection of each network. Measuring the protection is not enough, because the appearance of protection often hides complex and opaque dependencies.

Many networks seem "secure" not because they validate routes themselves, but because their upstream providers filter invalid announcements on their behalf. Also different networks could deploy ROV with different strategies and reliance. To uncover these hidden relationships and understand how ROV filtering actually propagates across the Internet, I developed **RScope** [2], the first measurement system capable of revealing the dependency chains and reliance patterns that determine how ROV protection emerges in practice. Using RScope, we found more than 67% of current protected networks are still

Together, RoVista and RScope provide the first comprehensive, Internet-scale understanding of how ROV protection actually emerges in today's routing ecosystem. These systems not only obtain a large-scale and accurate dataset for today's Internet routing security status, but also introduce a nuanced, data-driven view of how security is inherited, and enforced across the global Internet.

Data and AI drive security for Internet routing

Internet resource records—including WHOIS, ROAs, and IRR—were originally established to enable trust, validate ownership, and facilitate policy coordination in a decentralized ecosystem. Today, these records have become fundamental data sources for cybersecurity, widely used for measurement, attribution, and threat detection in both academia and industry. However, managing these datasets is inherently difficult: because they are maintained separately across different authorities and primarily rely on manual updates, they are frequently incomplete, outdated, or incorrect due to misconfiguration. **This unreliability makes it difficult for operators and researchers to rely on them for security-sensitive decisions.** My work addresses this problem by building automatic systems that detect misconfigurations, reconcile conflicting records, and use AI to infer missing or ambiguous information from Internet metadata.

My study on **ROA misconfigurations** [3] provides the first Internet-scale analysis of why RPKI-Invalid routes occur, showing that most arise from operational mistakes rather than attacks. By correlating RPKI objects with BGP updates and organizational metadata, we classify common error patterns mapped with real-world business models, and quantify their impact on data-plane that people ignore. This work not only answers "which is misconfiguration", but also answers why misconfiguration happens, what's the impact, and how we can prevent it. To improve the accuracy of legacy trust anchors, I also developed **IRRedicator** [4], which applies machine learning with RPKI-valid BGP observations to prune stale or conflicting records in the Internet Routing Registry (IRR).

Accuracy is not the only concern. Many real-world relationships on today's Internet are not recorded in public datasets and are missed by current inference methods, yet these hidden organizational and routing relationships are equally important for attributing incidents, validating configuration changes, and detecting security risks. To address this gap, I built **ASINT** [5], the first retrieval-augmented generation

(RAG) pipeline for routing security. ASINT integrates WHOIS, PeeringDB, RPKI, BGP, and public web data, and uses constrained LLM inference to construct high-coverage AS-to-organization mappings. This unified knowledge base improves attribution, reduces false positives in misconfiguration detection, and demonstrates how AI can safely interpret fragmented Internet metadata.

Securing routing protocol and policy

Routing security is fundamental to the stability and trustworthiness of the Internet and has obtained increasing attention. While understanding and identifying today's threat is important, **ensuring that routing protocols and software are secure and prepared for future challenges** is essential for achieving a long-term, sustainable solution. Although mechanisms such as RPKI were introduced to strengthen BGP, today's routing ecosystem remains fragile. Operational deployments often diverge from standards, new threat surfaces emerge as business models evolve, and legacy security infrastructures struggle to keep pace with Internet growth. Based on the measurement and data-driven threat analysis, my research **improves the security and performance of RPKI for new challenges including partial-deployment collateral damage, post-quantum crypto requirements, and new business models like IP leasing.**

Partial-deployment collateral damage: Although ROV were created to prevent origin hijacks, not all hijacks can be prevented unless every network on the Internet has deployed it which is far from where we are right now. To ensure the protection when RPKI are partially deployed across the Internet, I designed a new extension **ImpROV** [6]. ImpROV introduces new routing tables to "remember" non-protected neighbors and choose safest next-hop when hijack happens with only 2-3% performance overhead.

Post-quantum security: As the Internet transitions toward a post-quantum world, the RPKI ecosystem faces a critical challenge: its reliance on RSA becomes insecure, yet directly adopting PQ signatures dramatically increases certificate size, verification overhead, and repository churn. Existing PQC standards are not designed for a system as large and time-sensitive as RPKI. I propose a new RPKI standard **pqRPKI** [7], a practical, post-quantum–secure RPKI architecture that avoids these costs. pqRPKI uses a Merkle Tree Ladder combined with SPHINCS+ to shrink per-object metadata, improve validation performance, and reduce repository size to only 56% of original repositories—making PQ-secure routing not only feasible but operationally efficient.

IP Leasing business: The rise of commercial IP-leasing platforms has introduced a new and largely invisible type of routing relationship. On the other hand, legitimate ownership records and ROA access policy are still designed without considering the leasing business model. My research **points out the vulnerability when RPKI and IP leasing interact together** [8]: because leased resources often lack accurate RIR, IRR, and ROA records, attackers can obtain "legitimate-looking" prefixes and ASNs to bypass existing defenses. With real-world measurement study, my work shows a rogue lessor can hijack its leased-out prefixes without being detected by lessee and leasing brokers, evade public monitoring and leverage the natural vulnerability of RPKI RRDP protocol.

Public dataset, open source, and real-world impact.

The significance and impact of my work has been recognized through acceptance in top-tier conferences (Usenix Security, NDSS, IEEE S&P, and IMC). Beyond publications, my research provides longitudinal public datasets (like RoVista, RScope, ASINT) and open source softwares (like ImpROV and pqRPKI). My measurement platforms and data have been used/cited by governments agencies like FCC and NTIA, industries like Cloudflare and Comcast, and policy makers like ISOC. The explored vulnerability has raised discussion in IETF working groups and a new standard to replace RRDP protocol and mitigate security risk has been proposed by the SIDRops working groups.

Future Research

Looking ahead, my work will develop data-driven and AI-assisted approaches to improve the Internet's visibility, security, and resilience. My research aligns closely with computer science efforts that integrate networking, security, and data-driven systems to better understand and protect large-scale Internet infrastructure. I am particularly interested in collaborating with faculty whose work bridges networked systems, security, and large-scale measurement.

AI for routing security.

As the Internet's topology and business relationships grow more complex, distinguishing malicious routing events from ordinary misconfigurations has become increasingly difficult. Public datasets such as WHOIS, IRR, and RPKI provide only partial or outdated visibility, and many operational relationships such as resource transfers or leasing arrangements—are undocumented. This limited observability makes it challenging to infer hijacks, detect abuses, or understand the true intent behind routing changes.

To address these gaps, my future work will explore AI-driven approaches to routing security. By using retrieval-augmented generation, constrained LLM inference, and large-scale measurement data, AI systems can automatically retrieve, clean, and integrate fragmented Internet metadata into richer models of routing behavior. These models can support more accurate detection of hijacks, identification of suspicious resource use, and discovery of configuration anomalies, establishing AI-assisted analysis as a practical foundation for securing the increasingly complex global routing ecosystem.

Internet Data-driven cybersecurity threat detection.

As Internet services increasingly rely on complex, distributed infrastructures, many modern cyber threats manifest not through isolated vulnerabilities but through subtle, multi-layer behaviors observable only at Internet scale. Meanwhile, the rise of cloud hosting, rapid domain registration services, residential proxy networks, and IP-leasing markets has made it easier than ever for adversaries to obtain and rotate Internet resources—such as IP addresses and domain names—that traditionally served as stable identifiers for threat detection. This fluidity enables attackers to hide behind legitimate infrastructure, evade blocklists, and blend malicious behavior into normal operational noise.

My future research will focus on building Internet-scale, data-driven threat detection systems that fuse routing metadata, DNS/TLS signals, organizational intelligence, and AI-assisted inference to **identify**

these emerging abuse patterns beyond traditional Internet identifiers. By integrating multi-plane measurements with services-specific behavior inference methods, I aim to develop systems that can detect anomalous resource usage, uncover hidden infrastructure reuse, and provide early warnings of attacks that exploit the increasingly dynamic nature of Internet resources.

Mapping Internet's dependency and resilience

As the Internet grows more interconnected, disruptions in a single network can propagate across multiple providers, degrading connectivity, increasing latency, or isolating end users in unexpected ways. Yet real outages are unpredictable and difficult to study systematically, and many of the routing and business relationships that shape global resilience remain opaque. Without clear visibility into these hidden dependencies, it is challenging for operators, CDNs, and cloud providers to anticipate how failures in specific networks will impact their services or to design robust mitigation strategies.

To answer "**how outages impact the Internet**" and **help the Internet be prepared for outages**, my future work aims to map these dependencies and quantify the Internet's resilience using controlled, ethically safe outage-like experiments paired with large-scale data-plane measurements. By observing how connectivity and latency shift when specific paths become unavailable, this research will identify vulnerable regions, critical transit dependencies, and weakness points in anycast and peering structures. The resulting models and measurements will offer operators actionable guidance for improving infrastructure robustness, planning PoP deployment, and strengthening the reliability of Internet services under real-world failure scenarios.

# Reference (* denote equal contribution)

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